

MULTIPLICATIVE-ORDER PARAMETRIZED DIGRAPHS

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Abstract

Let $\mathcal{P}$ be a set of n distinct odd primes and m a positive integer. The *Multiplicative-Order Parametrized Digraph* $\Xi_m(\mathcal{P})$ has vertex set $\mathcal{P}$ and an edge (p, p') if and only if $p^m \equiv 1 \pmod{p'}$. The family $\Xi(\mathcal{P})$ obtained as m varies is therefore finite, and each member occurs on a set of exponents of positive density. We prove that $\Xi(\mathcal{P})$ consists of all $2^{n(n-1)}$ digraphs on $\mathcal{P}$, in which case $\mathcal{P}$ is called realizing, if and only if no multiplicative-order $\text{ord}_p(p')$ divides the least common multiple of the remaining ones, and that infinitely many realizing sets of every cardinality exist. We call $\mathcal{P}$ aligned when the divisors of the least common multiple of all the orders correspond bijectively to the digraphs on $\mathcal{P}$. Aligned sets containing a given prime with prescribed orders are finite in number and effectively computable, and we conjecture that none exists with $n \geq 4$, a statement reduced here to the case $n = 4$.

1. Introduction

Given a set $\mathcal{P} = \{p_1, \dots, p_n\}$ of n distinct odd primes and a positive integer m , the *multiplicative-order parametrized digraph* $\Xi_m(\mathcal{P})$ is the digraph with vertex set $\mathcal{P}$ having an edge from p_i to p_j exactly when $p_i^m \equiv 1 \pmod{p_j}$ as shown in Figure 1; we write $E_m(\mathcal{P})$ for its edge set, regarded as a set of ordered pairs (i, j) of distinct indices. Distinct primes are units modulo one another, so the multiplicative order $r_{ij} := \text{ord}_{p_j}(p_i)$ is defined for every such pair, and the condition defining an edge is precisely the divisibility $r_{ij} \mid m$, as shown in Fact 1. Varying m over the positive integers sweeps out the family $\Xi(\mathcal{P}) := \{\Xi_m(\mathcal{P}) : m \in \mathbb{N}\}$, and the question we address is which digraphs on the labelled vertex set $\mathcal{P}$ occur in it. Because each edge is switched on by a divisibility, the family depends on $\mathcal{P}$ only through the

multiset of orders $\{r_{ij}\}$, and it is finite: as proved in Corollary 1, two exponents congruent modulo $m_{\max} := \text{lcm}_{i \neq j} r_{ij}$ yield the same digraph, so it suffices to let m run over the divisors of $m_{\max}$, and $|\Xi(\mathcal{P})| \leq d(m_{\max})$ as shown in Proposition 1. At one extreme the family is as large as it can be, containing all $2^{n(n-1)}$ digraphs on $\mathcal{P}$; we call $\mathcal{P}$ a *realizing set* when this occurs.

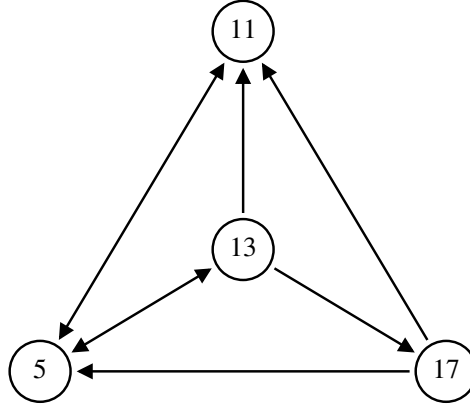


Figure 1: The digraph $\Xi_m(\mathcal{P})$ for $\mathcal{P} = \{5, 11, 13, 17\}$ and $m = 40$.

We write I_n for the set of $n(n-1)$ ordered pairs of distinct indices, so that the subsets of I_n are exactly the digraphs on the labelled vertex set $\mathcal{P}$, and d and ω for the number of divisors and the number of distinct prime factors. Congruent exponents have the same edge set, so each digraph of the family is realized on a periodic set of exponents and hence with a well-defined frequency: for $H \subseteq I_n$ let $\rho(H)$ be the natural density, which exists by Proposition 2, of $\{m \in \mathbb{N} : E_m(\mathcal{P}) = H\}$. Our first result characterizes the realizing sets intrinsically, by a condition on the orders alone.

Theorem A. *Let $\mathcal{P}$ be a set of n distinct odd primes. The following are equivalent.*

1. *(Lcm-independence) for every $e \in I_n$, $r_e \nmid \text{lcm}\{r_f : f \in I_n, f \neq e\}$.*
2. *the family satisfies $|\Xi(\mathcal{P})| = 2^{n(n-1)}$.*
3. *the density satisfies $\rho(H) > 0$ for every $H \subseteq I_n$.*

When they hold, $\omega(p_j - 1) \geq n - 1$ for every j , and $\omega(m_{\max}) \geq n(n-1)$.

Theorem A is composed of Theorem 1 together with Corollary 3. In the body the term *realizing set* is introduced by condition (1) rather than condition (2) in Definition 5; the equivalence just stated shows that the two descriptions agree. Its last two inequalities show that the condition is not met by an arbitrary set of

primes: the fundamental exponent of a realizing set of three primes is already at least $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 30030$, by Corollary 4. We prove that such sets nevertheless exist for every n , unconditionally.

Theorem B. *For every $n \geq 1$ there exist infinitely many sets $\mathcal{P}$ of n distinct odd primes with the following property: every $e \in I_n$ admits a prime q_e with $q_e \mid r_e$ and $q_e \nmid r_f$ for every $f \in I_n$ with $f \neq e$. Every such set satisfies the equivalent conditions of Theorem A. In particular, for every n there are infinitely many sets of n primes whose $\Xi(\mathcal{P})$ family contains all $2^{n(n-1)}$ digraphs on $\mathcal{P}$.*

This is Theorem 2 and Corollary 7; the proof combines Dirichlet's theorem on primes in arithmetic progressions, Kummer theory, and the Chebotarev density theorem, and uses no unproved hypothesis.

Lcm-independence decides which digraphs occur but it gives no information about how often they occur, nor about whether distinct divisors of the fundamental exponent produce distinct digraphs. Each of those two refinements is governed by an exact condition on the orders.

Theorem C. *Let $\mathcal{P}$ satisfy the equivalent conditions of Theorem A.*

1. *the orders r_e are pairwise coprime if and only if $m_{\max} = \prod_{e \in I_n} r_e$, and if and only if*

$$\rho(H) = \prod_{e \in H} \frac{1}{r_e} \prod_{e \notin H} \left(1 - \frac{1}{r_e}\right) \quad (H \subseteq I_n),$$

that is, if and only if the edges are independent events under ρ , the edge e being present with probability $1/r_e$.

2. *the orders r_e are pairwise distinct primes if and only if $|\Xi(\mathcal{P})| = d(m_{\max}) = 2^{n(n-1)}$, and if and only if the map $m' \mapsto E_{m'}(\mathcal{P})$ is an isomorphism of Boolean algebras from the lattice of divisors of $m_{\max}$ onto the power set of I_n .*

Theorem C is composed of Theorems 3 and 5. Sets with the property described in part (1) are called *pairwise coprime*, and form the class $\mathcal{R}_n^c$; those described in part (2) are called *aligned*, and form the class $\mathcal{R}_n^a$. In part (2) the standing hypothesis is redundant, each of the three conditions implying it. The two classes are rigid in a second, elementary way, through quadratic reciprocity; below $(\cdot)$ is the Legendre symbol.

Theorem D. *Let $n \geq 2$ and let $\mathcal{P}$ be a pairwise coprime set. Then at most one of the $n(n-1)$ orders is even, and at most two elements of $\mathcal{P}$ are congruent to 3 modulo 4. If exactly two are, say $p_a \equiv p_b \equiv 3 \pmod{4}$, then after possibly exchanging a and b the order r_{ab} is the unique even one and $(\frac{p_i}{p_j}) = 1$ for every $(i, j) \in I_n$ with $(i, j) \neq (a, b)$; if moreover $\mathcal{P}$ is aligned, then $r_{ab} = 2$ and $p_b \mid p_a + 1$.*

These are Proposition 7, Theorem 4, and Corollary 10. Aligned sets exist for $n = 2$: for $\mathcal{P} = \{3, 11\}$ the two orders are $\text{ord}_{11}(3) = 5$ and $\text{ord}_3(11) = 2$. None of larger cardinality is known. Section 4.3 proves two unconditional constraints on such a set, namely a divisibility forcing $\prod_{i \neq j} r_{ij} \mid p_j - 1$ and a confinement of p_j to the prime divisors of the single integer $\Phi_{r_{ij}}(p_i)$, and derives from a count of residues an upper bound for the expected number of aligned sets which is finite for $n \geq 4$ and vacuous for $n = 3$.

Conjecture E. There is no aligned set of cardinality $n \geq 4$.

Conjecture E is Conjecture 1. Both classes are closed under passage to subsets, so Conjecture E reduces to the single case $n = 4$. Whether an aligned set of cardinality 3 exists we leave open: the count just mentioned gives no information at that cardinality, in either direction. We do not know either whether there is a pairwise coprime set of cardinality $n \geq 3$; a negative answer would settle Conjecture E outright. For $n \geq 3$, accordingly, Theorems C and D describe classes that are not known to be nonempty, and Theorem B is the only one of the four that asserts existence.

Digraphs whose adjacency is decided by modular arithmetic have been studied in several forms, and it is worth saying at the outset how the present construction differs from them. Somer and Křížek [16, 17] fix a modulus N and an exponent k , take $\mathbb{Z}/N\mathbb{Z}$ as vertex set, and join a to b when $a^k \equiv b \pmod{N}$; the resulting power digraphs are studied one at a time. Arithmetic hypotheses on N and k are shown to characterize graph-theoretic properties such as symmetry or the number of components and the lengths of the cycles. The mechanism is the one used here, in that a congruence decides adjacency, but the data play interchanged roles. In $\Xi_m(\mathcal{P})$ the vertices are themselves the moduli: each p_j is the modulus for the $n - 1$ edges entering it and a base for the $n - 1$ edges leaving it, so that the arithmetic is reflexive and there is no ambient modulus distinct from the vertex set. Moreover, the object of study is not a single digraph but the family swept out by the scalar m with $\mathcal{P}$ held fixed.

A second precedent is the family of Paley graphs and Paley tournaments, whose vertex set is $\mathbb{F}_p$ and in which adjacency is decided by the quadratic character of the difference of two vertices. Graham and Spencer [4] proved that for every k the Paley tournament on $\mathbb{F}_p$ has the property that any k of its vertices are dominated by a common vertex, provided the prime $p \equiv 3 \pmod{4}$ is large enough in terms of k ; this answered a question of Schütte by an explicit arithmetic construction, replacing a counting argument. A prescribed combinatorial property is thereby secured by a choice of prime. Corollary 7 is of that kind, and the comparison isolates two differences. Adjacency in the Paley construction is governed by a single character of order two, whereas here the $n(n - 1)$ orders r_{ij} are independent parameters and the edges they control are switched separately; and universality is obtained there by choosing the ambient prime, so that the vertex set grows with the structure to

be embedded, whereas here the vertex set is fixed in advance and every digraph on it is realized by some exponent. A third construction of arithmetic origin, the prime distance graphs of Eggleton, Erdős, and Skilton [3], has $\mathbb{Z}$ as vertex set and joins integers whose difference is prime; there the defining condition is additive and the vertices carry no modular role.

The arithmetic input to our existence theorem places the paper in a second, older line of work, on the simultaneous prescription of multiplicative orders. Schinzel and Wójcik [14] proved that any two rationals outside $\{0, \pm 1\}$ generate the same subgroup of $\mathbb{F}_p^*$, and so have the same multiplicative order there, for infinitely many primes p . The corresponding question for r elements is known as the Schinzel–Wójcik problem and is open unconditionally; Wójcik [19] settled it under Schinzel’s Hypothesis H. Lenstra [9] supplied the Galois-theoretic framework on which the later work rests, and within that framework Pappalardi and Susa [12] obtained conditional densities, Järvinen [5] determined, under the generalized Riemann hypothesis, exactly which tuples of integers have equal orders modulo infinitely many primes, Just and Pollack [6] studied when one order exceeds another infinitely often, and Pollack [13] treated the analogue over number fields. Throughout that work the bases $a_1, \dots, a_k$ are fixed while the modulus p ranges over all primes, and the orders are prescribed as indices, in the form $\text{ord}_p(a_i) = (p-1)/\lambda_i$, so that the prescribed order grows with the modulus. Both features are reversed here. Bases and moduli are drawn from the same finite set $\mathcal{P}$, and the orders considered in Section 4 are prescribed as exact values, bounded independently of the primes involved. The second reversal is the more consequential: requiring $\text{ord}_{p_j}(p_i) = r$ for a fixed r forces $p_j \mid \Phi_r(p_i)$, which confines p_j to a finite set once p_i is fixed (this is Theorem 6 below), so that the density arguments of the works just cited have nothing to act on. This is not to say that no method reaches the reflexive case, only that these do not; we are not aware of any treatment of it in the literature.

Seen from that angle the questions of Section 4 lie on a scale of exact-order prescriptions of increasing severity. Forcing a prescribed prime to divide each order is the weakest of the three demands, and Theorem 2 meets it. Forcing every order to be as large as it can be, $r_{ij} = p_j - 1$, is the reflexive form of Artin’s primitive root conjecture, open already in its classical form; see Moree [10] for a survey. Forcing every order to be prime, with the primes distinct, is the opposite extreme, and Conjecture 1 asserts that for $n \geq 4$ it cannot be met at all. The last two share one obstruction, visible in the proof of Theorem 2: Kummer theory together with a Chebotarev argument produces primes for which a prescribed q divides an order, and gives no control over the cofactor.

Section 2 develops the parametrization: the reduction of an arbitrary exponent to a divisor of $m_{\max}$, the two bounds on $|\Xi(\mathcal{P})|$ that Theorems A and C later characterize as attained, and the densities $\rho(H)$. Section 3 proves Theorem A and the counting consequences of lcm-independence, and then Theorem B. Section 4 treats

the two stronger classes and proves Theorems C and D, ending with Conjecture E.

2. The Exponent Parametrization

Throughout, $\mathbb{N} := \{1, 2, 3, \dots\}$, and $\mathcal{P} = \{p_1, \dots, p_n\}$ is a set of $n \geq 1$ distinct odd primes; the letter n always denotes $|\mathcal{P}|$. We write

$$I_n := \{ (i, j) : 1 \leq i, j \leq n, i \neq j \}$$

for the set of $n(n-1)$ ordered pairs of distinct indices, and we use the letter e for a generic element of I_n . Subsets of I_n are identified with digraphs on the labelled vertex set $\mathcal{P}$. We write $\text{ord}_q(p)$ for the multiplicative order of p modulo q , and abbreviate $r_e = r_{ij} := \text{ord}_{p_j}(p_i)$ for $e = (i, j) \in I_n$. Finally $\text{lcm} \emptyset := 1$, and d , ω , and v_q denote respectively the number of divisors, the number of distinct prime factors, and the q -adic valuation of a positive integer; no graph invariant is denoted by d or ω in this paper.

Definition 1 (Multiplicative-order parametrized digraph). Let $m \in \mathbb{N}$. The *multiplicative-order parametrized digraph* $\Xi_m(\mathcal{P})$ is the digraph with vertex set $\mathcal{P}$ and edge set

$$E_m(\mathcal{P}) := \{ (i, j) \in I_n : p_i^m \equiv 1 \pmod{p_j} \}.$$

Fact 1 (Divisibility form of the edge relation). *For all $e = (i, j) \in I_n$ and all $m \in \mathbb{N}$,*

$$e \in E_m(\mathcal{P}) \quad \text{if and only if} \quad r_e \mid m.$$

Moreover, $r_e \mid p_j - 1$.

Proof. The primes $p_i \neq p_j$ are distinct, so p_i is a unit modulo p_j and r_e is its order in $(\mathbb{Z}/p_j\mathbb{Z})^*$. Hence, $p_i^m \equiv 1 \pmod{p_j}$ holds precisely when m is a multiple of r_e . The group $(\mathbb{Z}/p_j\mathbb{Z})^*$ has order $p_j - 1$, so $r_e \mid p_j - 1$ by Lagrange's theorem. $\square$

The single exponent m thus switches each potential edge on and off by a divisibility condition, and the entire parametrization is determined by the $n(n-1)$ integers r_e . The degenerate value $r_e = 1$ occurs exactly when $p_i \equiv 1 \pmod{p_j}$; such an edge belongs to every $E_m(\mathcal{P})$, so the empty digraph need not be realized at all.

Definition 2 (Family, order multiset, fundamental exponent). The *family* of $\mathcal{P}$ is the set of digraphs

$$\Xi(\mathcal{P}) := \{ \Xi_m(\mathcal{P}) : m \in \mathbb{N} \}.$$

The *order multiset* of $\mathcal{P}$ is $S(\mathcal{P}) := (r_e)_{e \in I_n}$, of cardinality $n(n-1)$ counted with multiplicity, and the *fundamental exponent* of $\mathcal{P}$ is

$$m_{\max}(\mathcal{P}) := \text{lcm } S(\mathcal{P}).$$

Lemma 1 (Reduction). *For every $m \in \mathbb{N}$,*

$$E_m(\mathcal{P}) = E_{\gcd(m, m_{\max}(\mathcal{P}))}(\mathcal{P}).$$

Proof. Fix $e \in I_n$. Then $r_e \mid m_{\max}(\mathcal{P})$ by Definition 2, so $r_e \mid m$ if and only if r_e divides $\gcd(m, m_{\max}(\mathcal{P}))$. Now apply Fact 1 in both directions. $\square$

Corollary 1. *Write $m_{\max} := m_{\max}(\mathcal{P})$. Then:*

1. *we have $\Xi_m(\mathcal{P}) = \Xi_{m'}(\mathcal{P})$ whenever $m \equiv m' \pmod{m_{\max}}$;*
2. *the family equals $\Xi(\mathcal{P}) = \{ \Xi_{m'}(\mathcal{P}) : m' \mid m_{\max} \}$;*
3. *the digraph $\Xi_m(\mathcal{P})$ is complete, in the sense that $E_m(\mathcal{P}) = I_n$, if and only if $m_{\max} \mid m$; in particular $m_{\max}$ is the least exponent realizing the complete digraph.*

Proof. For (1), congruent exponents have the same greatest common divisor with $m_{\max}$, so Lemma 1 gives equal edge sets. For (2), Lemma 1 exhibits every member of $\Xi(\mathcal{P})$ as $\Xi_{m'}(\mathcal{P})$ with $m' = \gcd(m, m_{\max}) \mid m_{\max}$, and each such m' is itself an admissible exponent. For (3), $E_m(\mathcal{P}) = I_n$ means $r_e \mid m$ for every e by Fact 1, that is, m is a common multiple of $S(\mathcal{P})$, which by the defining property of the least common multiple holds exactly when $m_{\max} \mid m$. Any positive multiple of $m_{\max}$ is at least $m_{\max}$. $\square$

Definition 3 (Minimal exponent of a digraph). For $H \subseteq I_n$ put $m_H := \text{lcm}\{r_e : e \in H\}$, so that $m_{\emptyset} = 1$ and $m_{I_n} = m_{\max}(\mathcal{P})$.

Lemma 2 (Upper closure). *Let $H \subseteq I_n$. Then*

$$\{m \in \mathbb{N} : H \subseteq E_m(\mathcal{P})\} = \{m \in \mathbb{N} : m_H \mid m\},$$

and $m_H \mid m_{\max}(\mathcal{P})$. If moreover $H \in \Xi(\mathcal{P})$, then $E_{m_H}(\mathcal{P}) = H$ and m_H is the least exponent realizing H ; in particular the map $H \mapsto m_H$ is injective on $\Xi(\mathcal{P})$.

Proof. By Fact 1, $H \subseteq E_m(\mathcal{P})$ means $r_e \mid m$ for every $e \in H$, which by the defining property of the least common multiple is $m_H \mid m$. Taking $H = I_n$ in the same equivalence and using Corollary 1(3) gives $m_H \mid m_{I_n} = m_{\max}(\mathcal{P})$ for every H , since m_H is a least common multiple of a subcollection.

Suppose now $H = E_m(\mathcal{P})$ for some m . The first part gives $m_H \mid m$ and $H \subseteq E_{m_H}(\mathcal{P})$. Conversely, if $e \in E_{m_H}(\mathcal{P})$ then $r_e \mid m_H \mid m$, so $e \in E_m(\mathcal{P}) = H$. Hence, $E_{m_H}(\mathcal{P}) = H$. Any m' with $E_{m'}(\mathcal{P}) = H$ satisfies $H \subseteq E_{m'}(\mathcal{P})$ and therefore $m_H \mid m'$, so $m_H \leq m'$. Injectivity follows because H is recovered from m_H as $E_{m_H}(\mathcal{P})$. $\square$

Proposition 1 (Two bounds). *For every $\mathcal{P}$,*

$$|\Xi(\mathcal{P})| \leq \min\left\{2^{n(n-1)}, d(m_{\max}(\mathcal{P}))\right\}.$$

Proof. All members of $\Xi(\mathcal{P})$ share the vertex set $\mathcal{P}$, on which there are $2^{n(n-1)}$ digraphs. The second bound is Corollary 1(2), the number of divisors of $m_{\max}(\mathcal{P})$ being $d(m_{\max}(\mathcal{P}))$. $\square$

The two bounds of Proposition 1 are independent, and the classes of Section 4 are precisely the conditions under which each of them, and then both simultaneously, are attained.

The parametrization carries more information than the list of realized digraphs: by Corollary 1(1) each digraph is realized on a periodic set of exponents, and therefore with a well-defined frequency.

Definition 4 (Realization density). For $H \subseteq I_n$, the *realization density* of H is the natural density $\rho(H)$ of the set $\{m \in \mathbb{N} : E_m(\mathcal{P}) = H\}$.

Proposition 2 (Realization density). *The density $\rho(H)$ exists for every $H \subseteq I_n$, and*

$$\rho(H) = \sum_{H' \supseteq H} \frac{(-1)^{|H' \setminus H|}}{m_{H'}}, \quad (1)$$

the sum being over the subsets H' of I_n containing H . Moreover, $\sum_{H \subseteq I_n} \rho(H) = 1$, and $\rho(H) > 0$ if and only if $H \in \Xi(\mathcal{P})$.

Proof. By Corollary 1(1) the set $\{m : E_m(\mathcal{P}) = H\}$ is a union of residue classes modulo $m_{\max}(\mathcal{P})$; a periodic set has a natural density, equal to the proportion of classes it occupies, so $\rho(H)$ exists. The same applies to $\{m : H \subseteq E_m(\mathcal{P})\}$, which by Lemma 2 is the set of multiples of m_H and hence has density $1/m_H$. Sorting those exponents by the exact value of $E_m(\mathcal{P})$ gives

$$\frac{1}{m_H} = \sum_{H' \supseteq H} \rho(H') \quad (H \subseteq I_n),$$

a finite identity of densities, since the events are disjoint and finitely many. Möbius inversion over the Boolean lattice of subsets of I_n , with Möbius function $\mu(H, H') = (-1)^{|H' \setminus H|}$, yields Equation (1). Taking $H = \emptyset$ in the displayed identity and using $m_{\emptyset} = 1$ gives $\sum_{H'} \rho(H') = 1$. Finally a periodic set has positive density if and only if it is nonempty, and it is nonempty exactly when H is realized. $\square$

The edge sets are compatible with one arithmetic operation on the exponent for every set of primes, without any further hypothesis.

Corollary 2 (Intersection of edge sets). *For all $m, m' \in \mathbb{N}$,*

$$E_m(\mathcal{P}) \cap E_{m'}(\mathcal{P}) = E_{\gcd(m, m')}(\mathcal{P}),$$

and $E_m(\mathcal{P}) \subseteq E_{m'}(\mathcal{P})$ whenever $m \mid m'$.

Proof. Fix $e \in I_n$. By Fact 1, the pair e lies in $E_m(\mathcal{P}) \cap E_{m'}(\mathcal{P})$ if and only if r_e divides both m and m' , which by the defining property of the greatest common divisor holds if and only if $r_e \mid \gcd(m, m')$, that is, if and only if $e \in E_{\gcd(m, m')}(\mathcal{P})$. If $m \mid m'$, then $\gcd(m, m') = m$ and the displayed identity gives $E_m(\mathcal{P}) \subseteq E_{m'}(\mathcal{P})$. $\square$

3. Realizing Sets

The classes of prime sets studied below are defined solely by the divisibility relations among the orders r_e ; none of them refers to the exponent. The weakest of them is the exact condition under which the first bound of Proposition 1 is attained.

Definition 5 (Realizing set). The set $\mathcal{P}$ is a *realizing set*, written $\mathcal{P} \in \mathcal{R}_n$, if it satisfies the *lcm-independence*: for every $e \in I_n$,

$$r_e \nmid \text{lcm}\{r_f : f \in I_n, f \neq e\}.$$

Lemma 3 (Valuation form of lcm-independence). *Let $e \in I_n$. Then*

$$r_e \nmid \text{lcm}\{r_f : f \neq e\}$$

if and only if there is a prime q with

$$v_q(r_e) > v_q(r_f) \quad \text{for all } f \in I_n, f \neq e.$$

Accordingly, $\mathcal{P} \in \mathcal{R}_n$ if and only if every $e \in I_n$ admits such a prime, which we call a witness prime for e .

Proof. For any prime q , the valuation of a least common multiple is the maximum of the valuations, so

$$v_q(\text{lcm}\{r_f : f \neq e\}) = \max_{f \neq e} v_q(r_f).$$

A positive integer divides another precisely when its valuation is at most that of the other at every prime. Hence, r_e divides the above least common multiple if and only if $v_q(r_e) \leq \max_{f \neq e} v_q(r_f)$ for every prime q . Negating, r_e fails to divide it if and only if some prime q satisfies $v_q(r_e) > \max_{f \neq e} v_q(r_f)$, which is the strict valuation dominance in the statement. The last assertion is Definition 5 applied at every $e \in I_n$. $\square$

Theorem 1 (Characterization of realizing sets). *The following are equivalent.*

1. *we have $\mathcal{P} \in \mathcal{R}_n$;*
2. *the family satisfies $|\Xi(\mathcal{P})| = 2^{n(n-1)}$, that is, every digraph on the labelled vertex set $\mathcal{P}$ is realized by some exponent;*
3. *the density satisfies $\rho(H) > 0$ for every $H \subseteq I_n$.*

Proof. The equivalence of (2) and (3) is immediate from Proposition 2: a subset H satisfies $\rho(H) > 0$ exactly when $H \in \Xi(\mathcal{P})$, and $\Xi(\mathcal{P})$ consists of all $2^{n(n-1)}$ subsets of I_n exactly when this holds for every H .

Condition (1) implies (2). Let $H \subseteq I_n$ be arbitrary. By Lemma 2 we have $H \subseteq E_{m_H}(\mathcal{P})$. For the reverse inclusion let $e \notin H$. Every order entering m_H is of the form r_f with $f \neq e$, whence $m_H \mid \text{lcm}\{r_f : f \neq e\}$. Lcm-independence at e gives $r_e \nmid \text{lcm}\{r_f : f \neq e\}$, so $r_e \nmid m_H$, and $e \notin E_{m_H}(\mathcal{P})$ by Fact 1. Hence, $E_{m_H}(\mathcal{P}) = H$ and $H \in \Xi(\mathcal{P})$.

Condition (2) implies (1). Suppose lcm-independence fails at some $e \in I_n$, so that $r_e \mid \text{lcm}\{r_f : f \neq e\}$, and put $H := I_n \setminus \{e\}$. The right-hand side is exactly m_H , so $r_e \mid m_H$ and $e \in E_{m_H}(\mathcal{P})$ by Fact 1. Were H realized, Lemma 2 would give $E_{m_H}(\mathcal{P}) = H$, which excludes e . Hence, $H \notin \Xi(\mathcal{P})$ and $|\Xi(\mathcal{P})| < 2^{n(n-1)}$. $\square$

Example 1. Let $\mathcal{P} = \{5, 7\}$, with the indices assigned in this order. Then $r_{12} = \text{ord}_7(5) = 6$ and $r_{21} = \text{ord}_5(7) = 4$, so $m_{\max}(\mathcal{P}) = 12$. Neither order divides the other, so $\mathcal{P} \in \mathcal{R}_2$; equivalently, all four digraphs on two labelled vertices occur, with densities

$$\rho(\emptyset) = \frac{2}{3}, \quad \rho(\{(1, 2)\}) = \frac{1}{12}, \quad \rho(\{(2, 1)\}) = \frac{1}{6}, \quad \rho(I_2) = \frac{1}{12},$$

in accordance with Equation (1). The witness primes are 3 for $(1, 2)$ and 2 for $(2, 1)$, the latter by $v_2(4) = 2 > 1 = v_2(6)$.

Example 2. Let $\mathcal{P} = \{4421, 4423, 5153\}$, with the indices assigned in this order. The six orders and their factorizations are

$$\begin{aligned} r_{12} &= 1474 = 2 \cdot 11 \cdot 67, & r_{13} &= 644 = 2^2 \cdot 7 \cdot 23, & r_{21} &= 340 = 2^2 \cdot 5 \cdot 17, \\ r_{23} &= 368 = 2^4 \cdot 23, & r_{31} &= 221 = 13 \cdot 17, & r_{32} &= 201 = 3 \cdot 67, \end{aligned}$$

and every ordered pair admits exactly one witness prime in the sense of Lemma 3, namely 11, 7, 5, 2, 13, and 3 in the order just displayed; the witness for $(2, 3)$ is the prime 2, since $v_2(r_{23}) = 4$ exceeds the 2-adic valuation of each of the other five orders. Hence, $\mathcal{P} \in \mathcal{R}_3$ by Theorem 1, and all $2^6 = 64$ digraphs on the labelled vertex set occur; Figure 2 displays the 16 isomorphism types they represent. The fundamental exponent is $m_{\max}(\mathcal{P}) = 2^4 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 67 = 6\,293\,567\,280$, which has 1280 divisors, so the first bound of Proposition 1 is attained here and the second is not. The complete digraph is the rarest member of the family, of density $1/m_{\max}(\mathcal{P})$, whereas the empty digraph has density greater than 0.98.

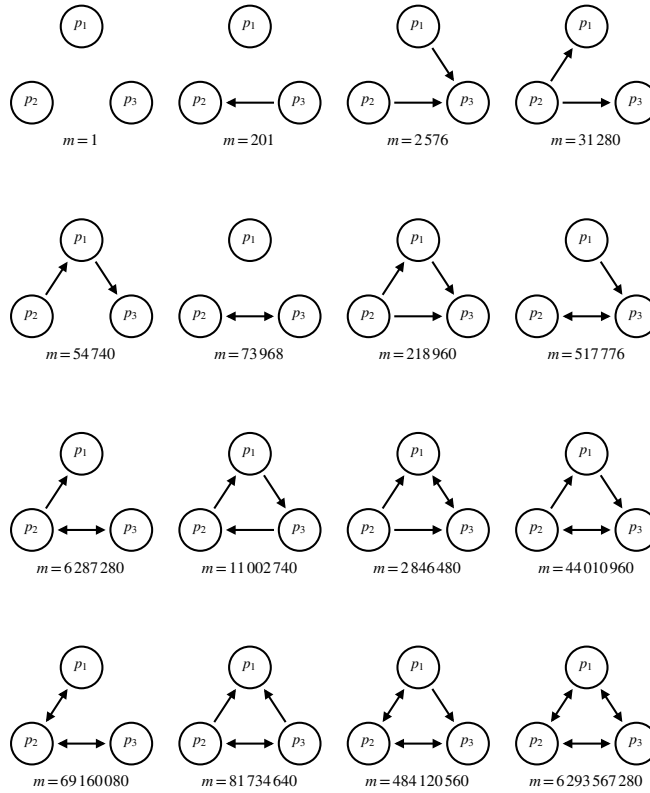


Figure 2: The non-isomorphic digraphs of the family $\Xi(\mathcal{P})$ for the realizing set $\mathcal{P} = \{4421, 4423, 5153\} = \{p_1, p_2, p_3\}$ as m varies through the divisors of $m_{\max}$.

The order attached to an ordered pair of distinct primes depends only on that pair and not on any ambient set, so the lcm-independence can be compared across a set and its subsets. It is inherited downwards.

Proposition 3 (Heredity). *Let $\mathcal{P} \in \mathcal{R}_n$ and let $\mathcal{P}' \subseteq \mathcal{P}$ with $|\mathcal{P}'| = s \geq 1$. Then $\mathcal{P}' \in \mathcal{R}_s$, and consequently $|\Xi(\mathcal{P}')| = 2^{s(s-1)}$.*

Proof. Let $I' \subseteq I_n$ be the set of ordered pairs of distinct indices whose two entries both correspond to elements of $\mathcal{P}'$; the orders of $\mathcal{P}'$ are the r_e with $e \in I'$. Fix $e \in I'$. Since $\{r_f : f \in I', f \neq e\} \subseteq \{r_f : f \in I_n, f \neq e\}$, the least common multiple of the former divides that of the latter. By lcm-independence in $\mathcal{P}$ the order r_e does not divide the latter, hence it does not divide the former. As $e \in I'$ was arbitrary, $\mathcal{P}'$ satisfies the lcm-independence. The last assertion is Theorem 1 applied to $\mathcal{P}'$. $\square$

The lcm-independence is stated pair by pair, but it forces a global constraint on the orders, which the remaining results of this section extract.

Proposition 4 (Witness primes are globally distinct). *Let $\mathcal{P} \in \mathcal{R}_n$ and, for each $e \in I_n$, let q_e be a witness prime for e . Then $q_e \neq q_f$ whenever $e \neq f$. In particular $\mathcal{P}$ admits $n(n-1)$ pairwise distinct witness primes, one for each ordered pair, and $q_e \mid r_e$ for every e .*

Proof. Suppose $q_e = q_f =: q$ for some $e \neq f$. Applying Lemma 3 to e with the pair f , and to f with the pair e , gives $v_q(r_e) > v_q(r_f)$ and $v_q(r_f) > v_q(r_e)$ simultaneously, which is impossible. The choice of one witness prime per pair therefore produces $n(n-1)$ distinct primes. Finally $v_{q_e}(r_e) > v_{q_e}(r_f) \geq 0$ forces $v_{q_e}(r_e) \geq 1$, that is, $q_e \mid r_e$. $\square$

Corollary 3 (Capacity of the moduli). *Let $\mathcal{P} \in \mathcal{R}_n$. Then*

$$\omega(p_j - 1) \geq n - 1 \quad \text{for every } j, \quad \text{and} \quad \omega(m_{\max}(\mathcal{P})) \geq n(n - 1).$$

Proof. Fix j and choose a witness prime q_{ij} for each of the $n-1$ pairs (i, j) with $i \neq j$. By Proposition 4 these are pairwise distinct, and $q_{ij} \mid r_{ij} \mid p_j - 1$ by Fact 1; hence $p_j - 1$ has at least $n-1$ distinct prime factors. Ranging over all of I_n instead produces $n(n-1)$ pairwise distinct primes, each dividing its own order and hence dividing $m_{\max}(\mathcal{P})$. $\square$

Corollary 4 (Primorial lower bound). *Let $\mathcal{P} \in \mathcal{R}_n$ and put $k := n(n-1)$. Then $m_{\max}(\mathcal{P})$ is at least the product of the k smallest primes, and consequently*

$$\log m_{\max}(\mathcal{P}) \geq (1 + o(1)) k \log k = (1 + o(1)) n(n-1) \log(n(n-1)) \quad (n \rightarrow \infty).$$

Proof. By Corollary 3 the integer $m_{\max}(\mathcal{P})$ is divisible by k pairwise distinct primes, so it is at least their product, which in turn is at least the product $\ell_1 \cdots \ell_k$ of the k smallest primes. Taking logarithms, $\log m_{\max}(\mathcal{P}) \geq \sum_{i \leq k} \log \ell_i = \theta(\ell_k)$, where θ is the Chebyshev function. By the prime number theorem $\theta(x) = (1 + o(1))x$ and $\ell_k = (1 + o(1)) k \log k$ [2], whence $\theta(\ell_k) = (1 + o(1)) k \log k$. $\square$

3.1. Existence

We now show that $\mathcal{R}_n$ is infinite for every n . The primes are selected one at a time, and at each step a Chebotarev argument in a Kummer extension forces the orders created by the new prime to be divisible by prescribed auxiliary primes and by no others. The construction attaches to each ordered pair a prime that divides that pair's order and no other, a condition strictly stronger than the strict valuation dominance of Lemma 3, since it requires valuation zero elsewhere rather than merely smaller valuation.

Definition 6 (Network of auxiliary primes). Let $J \subseteq I_n$. A *network on J* is a family $\{q_{ij} : (i, j) \in J\}$ of pairwise distinct odd primes satisfying the *non-resonance* requirement

$$q \nmid q' - 1 \quad \text{for all network primes } q, q'. \quad (2)$$

We call q_{ij} the *intended private prime* of the pair (i, j) , and put

$$R_j := \prod_{i : (i, j) \in J} q_{ij}, \quad R := \prod_{j=1}^n R_j = \prod_{(i, j) \in J} q_{ij}.$$

Since the q_{ij} are pairwise distinct, the R_j are pairwise coprime.

A network exists on any J : enumerate the required primes in increasing order and, at each step, choose the next one in a residue class avoiding $1 \bmod q$ for every q already chosen, which is possible by Dirichlet's theorem [2]; a prime q larger than q' cannot divide $q' - 1 < q$, so Equation (2) holds in both directions. Non-resonance is exactly what makes $[\mathbb{Q}(\zeta_{R_j}) : \mathbb{Q}] = \prod_i (q_{ij} - 1)$ coprime to each q_{ij} , a hypothesis of Lemma 5 that is not automatic, as $3 \mid 7 - 1$ shows.

Lemma 4 (Power-residue criterion). *Let p be an odd prime, let q be a prime with $v_q(p-1) = 1$, and let $x \in (\mathbb{Z}/p\mathbb{Z})^*$. Then x is a q -th power residue modulo p if and only if $q \nmid \text{ord}_p(x)$.*

Proof. The group $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic of order $p-1$, so x is a q -th power if and only if $x^{(p-1)/q} = 1$, that is, if and only if $\text{ord}_p(x) \mid (p-1)/q$. Since $v_q(p-1) = 1$, a divisor of $p-1$ divides $(p-1)/q$ precisely when q does not divide it. $\square$

Lemma 5 (Linear independence in $\mathbb{k}^*/(\mathbb{k}^*)^q$). *Let q be an odd prime and let $\mathbb{k}$ be a number field with $\gcd([\mathbb{k} : \mathbb{Q}], q) = 1$. Let $p_1, \dots, p_s$ be distinct rational primes, all different from q . Then the classes of $p_1, \dots, p_s$ are $\mathbb{F}_q$ -linearly independent in $\mathbb{k}^*/(\mathbb{k}^*)^q$.*

Proof. Suppose $a_1, \dots, a_s \in \{0, \dots, q-1\}$ satisfy $\prod_{i=1}^s p_i^{a_i} = \alpha^q$ for some $\alpha \in \mathbb{k}^*$. Write $D := [\mathbb{k} : \mathbb{Q}]$ and apply the norm $N_{\mathbb{k}/\mathbb{Q}}$:

$$\prod_{i=1}^s p_i^{a_i D} = N_{\mathbb{k}/\mathbb{Q}}(\alpha^q) = N_{\mathbb{k}/\mathbb{Q}}(\alpha)^q.$$

Set $x := N_{\mathbb{k}/\mathbb{Q}}(\alpha) \in \mathbb{Q}^*$. Comparing p_i -adic valuations in $\mathbb{Q}^*$ gives $a_i D = q \cdot v_{p_i}(x)$, so $q \mid a_i D$. As $\gcd(q, D) = 1$ we get $q \mid a_i$, and $0 \leq a_i < q$ forces $a_i = 0$. $\square$

Corollary 5 (Disjointness of the Kummer extensions). *In the setting of Lemma 5, assume moreover $\zeta_q \in \mathbb{k}$. Then $[\mathbb{k}(p_1^{1/q}, \dots, p_s^{1/q}) : \mathbb{k}] = q^s$ and*

$$\text{Gal}\left(\mathbb{k}(p_1^{1/q}, \dots, p_s^{1/q})/\mathbb{k}\right) \cong (\mathbb{Z}/q\mathbb{Z})^s.$$

Proof. By Kummer theory [8] the extension $\mathbb{k}(\Delta^{1/q})/\mathbb{k}$ generated by a subgroup $\Delta \leq \mathbb{k}^*/(\mathbb{k}^*)^q$ has Galois group dual to Δ , of order $|\Delta|$. Lemma 5 gives $\dim_{\mathbb{F}_q} \Delta = s$, so $|\Delta| = q^s$. $\square$

Proposition 5 (Intersection with the cyclotomic field). *Let $\{q_{ij}\}$ be a network on J , fix t with $1 \leq t \leq n$, and let $p_1, \dots, p_{t-1}$ be distinct rational primes all larger than every network prime. Put*

$$\mathbb{k}_t := \mathbb{Q}(\zeta_{R_t}), \quad F_t := \mathbb{k}_t \left(\left\{ p_b^{1/q} : 1 \leq b \leq t-1, q \mid R_t \right\} \right).$$

Then:

1. *we have $\text{Gal}(F_t/\mathbb{k}_t) \cong \prod_{q \mid R_t} (\mathbb{Z}/q\mathbb{Z})^{t-1}$;*
2. *the maximal subextension of $F_t/\mathbb{Q}$ abelian over $\mathbb{Q}$ is exactly $\mathbb{k}_t$;*
3. *we have $F_t \cap \mathbb{Q}(\zeta_M) = \mathbb{k}_t$ for every positive integer M with $R_t \mid M$.*

Proof. If $t = 1$ the index set of Kummer generators is empty, so $F_1 = \mathbb{k}_1$. Then $\text{Gal}(F_1/\mathbb{k}_1)$ is trivial and part (1) reads as the empty product; part (2) holds because $\mathbb{k}_1/\mathbb{Q}$ is abelian; and part (3) is immediate from $\mathbb{k}_1 \subseteq \mathbb{Q}(\zeta_M)$, which follows from $R_1 \mid M$. We assume $t \geq 2$ for the remainder of the proof.

(1). Fix $q \mid R_t$. We have $[\mathbb{k}_t : \mathbb{Q}] = \varphi(R_t) = \prod_{q' \mid R_t} (q' - 1)$, which is coprime to q by the non-resonance requirement of Equation (2), and $\zeta_q \in \mathbb{k}_t$. The primes $p_1, \dots, p_{t-1}$ are distinct and different from q , so Corollary 5 applied with $\mathbb{k} = \mathbb{k}_t$ gives $\text{Gal}(F_t^{(q)}/\mathbb{k}_t) \cong (\mathbb{Z}/q\mathbb{Z})^{t-1}$, where $F_t^{(q)} := \mathbb{k}_t(p_1^{1/q}, \dots, p_{t-1}^{1/q})$. Enumerate the distinct primes dividing R_t as $q_1, \dots, q_h$ and set $N_0 := \mathbb{k}_t$ and $N_l := N_{l-1} \cdot F_t^{(q_l)}$ for $1 \leq l \leq h$, so that $N_h = F_t$. Each $F_t^{(q_l)}/\mathbb{k}_t$ is Galois of degree q_l^{t-1} . We show by induction on l that $[N_l : \mathbb{k}_t] = \prod_{u \leq l} q_u^{t-1}$ and $\text{Gal}(N_l/\mathbb{k}_t) \cong \prod_{u \leq l} (\mathbb{Z}/q_u\mathbb{Z})^{t-1}$. The case $l = 0$ is trivial. Assume the claim for $l - 1$. Then $[N_{l-1} : \mathbb{k}_t] = \prod_{u < l} q_u^{t-1}$ is coprime to $[F_t^{(q_l)} : \mathbb{k}_t] = q_l^{t-1}$. The compositum of two finite Galois extensions of $\mathbb{k}_t$ has degree over $\mathbb{k}_t$ dividing the product of their degrees and divisible by each of them; as the two degrees are coprime, this forces $[N_l : \mathbb{k}_t] = [N_{l-1} : \mathbb{k}_t] \cdot [F_t^{(q_l)} : \mathbb{k}_t]$, hence $N_{l-1} \cap F_t^{(q_l)} = \mathbb{k}_t$ and restriction gives $\text{Gal}(N_l/\mathbb{k}_t) \cong \text{Gal}(N_{l-1}/\mathbb{k}_t) \times \text{Gal}(F_t^{(q_l)}/\mathbb{k}_t)$. Taking $l = h$ proves (1).

(2). Put $G := \text{Gal}(F_t/\mathbb{Q})$ and $\mathcal{H} := \text{Gal}(F_t/\mathbb{k}_t)$. The extension $F_t/\mathbb{Q}$ is Galois, being the splitting field of $\prod_{b,q} (X^q - p_b)$ together with $X^{R_t} - 1$. Since $G/\mathcal{H} \cong \text{Gal}(\mathbb{k}_t/\mathbb{Q})$ is abelian, $[G, G] \subseteq \mathcal{H}$. We show $[G, G] = \mathcal{H}$; as the fixed field of $[G, G]$ is the maximal subextension abelian over $\mathbb{Q}$, this gives the claim.

Write $\mathcal{H} = \prod_{q \mid R_t} \mathcal{H}_q$ with $\mathcal{H}_q \cong (\mathbb{Z}/q\mathbb{Z})^{t-1}$ the factor from part (1), that is, $\mathcal{H}_q$ consists of the elements of $\mathcal{H}$ acting trivially on $p_b^{1/q'}$ for all $q' \neq q$ and all b . Since $\mathbb{k}_t/\mathbb{Q}$ is Galois, $\mathcal{H} \trianglelefteq G$. Let $\tau \in \mathcal{H}_q$, let $b \leq t-1$, and fix a q -th root $p_b^{1/q} \in F_t$; write $\tau(p_b^{1/q}) = \zeta p_b^{1/q}$ with $\zeta \in \mu_q$. Let $\sigma \in G$. As $p_b \in \mathbb{Q}$, the automorphism σ^{-1}

permutes the q -th roots of p_b , say $\sigma^{-1}(p_b^{1/q}) = \zeta' p_b^{1/q}$ with $\zeta' \in \mu_q$; applying σ gives $\sigma(p_b^{1/q}) = \sigma(\zeta')^{-1} p_b^{1/q}$. Moreover, $\mu_q \subseteq \mathbb{k}_t$ and τ fixes $\mathbb{k}_t$ pointwise, so $\tau(\zeta') = \zeta'$. Therefore,

$$\sigma\tau\sigma^{-1}(p_b^{1/q}) = \sigma(\tau(\zeta' p_b^{1/q})) = \sigma(\zeta' \zeta p_b^{1/q}) = \sigma(\zeta') \sigma(\zeta) \sigma(p_b^{1/q}) = \sigma(\zeta) p_b^{1/q}.$$

The same computation with q replaced by a prime $q' \mid R_t$ with $q' \neq q$ gives $\sigma\tau\sigma^{-1}(p_b^{1/q'}) = p_b^{1/q'}$, since τ acts trivially on $p_b^{1/q'}$. Hence, $\sigma\tau\sigma^{-1}$ lies in $\mathcal{H}$ and acts trivially on every $p_b^{1/q'}$ with $q' \neq q$, that is, $\sigma\tau\sigma^{-1} \in \mathcal{H}_q$; so $\mathcal{H}_q \trianglelefteq G$. Writing $\sigma(\zeta) = \zeta^{\chi_q(\sigma)}$, where $\chi_q : \text{Gal}(\mathbb{k}_t/\mathbb{Q}) \rightarrow (\mathbb{Z}/q\mathbb{Z})^*$ is the mod- q cyclotomic character and σ acts on μ_q through its restriction to $\mathbb{k}_t$, the displayed identity reads

$$\sigma\tau\sigma^{-1} = \tau^{\chi_q(\sigma)} \quad (\tau \in \mathcal{H}_q, \sigma \in G).$$

By the Chinese remainder theorem $\text{Gal}(\mathbb{k}_t/\mathbb{Q}) \cong \prod_{q' \mid R_t} (\mathbb{Z}/q'\mathbb{Z})^*$, so χ_q is surjective. As q is odd we may pick σ with $c := \chi_q(\sigma) \neq 1$; then $[\sigma, \tau] = \tau^{c-1}$ for every $\tau \in \mathcal{H}_q$, and $c-1$ is invertible modulo q , so $\mathcal{H}_q = \{\tau^{c-1} : \tau \in \mathcal{H}_q\} \subseteq [G, G]$. Taking the product over q gives $\mathcal{H} \subseteq [G, G]$, hence $[G, G] = \mathcal{H}$.

(3). Let $\alpha \in F_t \cap \mathbb{Q}(\zeta_M)$. The extension $\mathbb{Q}(\zeta_M)/\mathbb{Q}$ is abelian, so $\mathbb{Q}(\alpha)$ is a subextension of $F_t/\mathbb{Q}$ abelian over $\mathbb{Q}$; by (2), $\alpha \in \mathbb{k}_t$. Conversely, $\mathbb{k}_t \subseteq F_t$ by construction, and $\mathbb{k}_t \subseteq \mathbb{Q}(\zeta_M)$ because $R_t \mid M$. $\square$

Corollary 6 (Conjugation stability of the support conditions). *Keep the notation of Proposition 5 and assume $t \geq 2$. For $\tau \in \text{Gal}(F_t/\mathbb{k}_t)$, an index $b \leq t-1$ and a prime $q \mid R_t$, the statement*

$$\tau(p_b^{1/q}) = p_b^{1/q}$$

does not depend on the choice of the q -th root $p_b^{1/q} \in F_t$. Given a set T of pairs (b, q) with $1 \leq b \leq t-1$ and $q \mid R_t$, put

$$\mathcal{D}_T := \{ \tau \in \text{Gal}(F_t/\mathbb{k}_t) : \tau(p_b^{1/q}) \neq p_b^{1/q} \text{ if and only if } (b, q) \in T \}.$$

Then $\mathcal{D}_T$ is nonempty and stable under conjugation by $\text{Gal}(F_t/\mathbb{Q})$.

Proof. Two q -th roots of p_b in F_t differ by a factor in $\mu_q \subseteq \mathbb{k}_t$, and τ fixes $\mathbb{k}_t$ pointwise; hence τ fixes one of them if and only if it fixes all of them, which is the first assertion.

Write $\mathcal{H} := \text{Gal}(F_t/\mathbb{k}_t)$ and $G := \text{Gal}(F_t/\mathbb{Q})$, and let $\mathcal{H} = \prod_{q \mid R_t} \mathcal{H}_q$ be the decomposition of the proof of Proposition 5(2), where $\mathcal{H}_q \cong (\mathbb{Z}/q\mathbb{Z})^{t-1}$ by Proposition 5(1), the coordinate of $\tau \in \mathcal{H}_q$ at b being the exponent $a_b(\tau) \in \mathbb{Z}/q\mathbb{Z}$ determined by $\tau(p_b^{1/q}) = \zeta_q^{a_b(\tau)} p_b^{1/q}$. In these coordinates, $\tau \in \mathcal{H}$ moves $p_b^{1/q}$ precisely when the b -th coordinate of its component $\tau_q \in \mathcal{H}_q$ is nonzero, so $\mathcal{D}_T$ is the set of $\tau \in \mathcal{H}$ whose coordinate vector has support exactly T . Choosing the coordinate 1 at each pair of T and 0 elsewhere exhibits an element of $\mathcal{D}_T$, which is therefore nonempty.

Let $\sigma \in G$ and $\tau \in \mathcal{D}_T$, and write $\tau = \prod_q \tau_q$ with $\tau_q \in \mathcal{H}_q$. Each $\mathcal{H}_q$ is normal in G and, by the identity displayed in the proof of Proposition 5(2),

$$\sigma \tau_q \sigma^{-1} = \tau_q^{\chi_q(\sigma)},$$

where $\chi_q(\sigma) \in (\mathbb{Z}/q\mathbb{Z})^*$. Hence, $\sigma \tau \sigma^{-1} = \prod_q \tau_q^{\chi_q(\sigma)}$, and in the coordinates above the passage from τ_q to $\tau_q^{\chi_q(\sigma)}$ is multiplication of the coordinate vector by the invertible scalar $\chi_q(\sigma)$. Such a multiplication preserves the set of vanishing coordinates, so $\sigma \tau \sigma^{-1}$ has the same support T as τ , that is, $\sigma \tau \sigma^{-1} \in \mathcal{D}_T$. $\square$

The primes are selected one at a time, and at each stage the available control is of two different kinds. Modulo an *earlier* prime p_b the order $\text{ord}_{p_b}(p_t)$ can be pinned exactly, because there p_t is the unknown and a congruence determines it. Modulo the *new* prime p_t only divisibility statements about $\text{ord}_{p_t}(p_b)$ are available, because there p_t is the modulus and Chebotarev sees only Frobenius classes. The next lemma packages both, together with the auxiliary conditions needed to iterate.

Lemma 6 (Selection). *Let $\{q_{ij}\}$ be a network on $J \subseteq I_n$ and let $1 \leq t \leq n$. Suppose distinct odd primes $p_1, \dots, p_{t-1}$, each larger than every network prime, have been chosen so that for every $b < t$:*

- (A_b) $v_q(p_b - 1) = 1$ for every prime $q \mid R_b$;
- (B_b) $p_b \not\equiv 1 \pmod{q}$ for every network prime $q \nmid R_b$.

Let there be given:

- an integer $W \geq 1$ with $\gcd(W, R \prod_{b < t} p_b) = 1$ and a class $w \in (\mathbb{Z}/W\mathbb{Z})^*$;
- for each $b < t$, a class $c_b \in (\mathbb{Z}/p_b\mathbb{Z})^*$;
- for each $b < t$ and each prime $q \mid R_t$, a bit $\varepsilon_{b,q} \in \{0, 1\}$.

Then there exist infinitely many primes p_t , exceeding every network prime and every p_b with $b < t$, satisfying

- (A) $v_q(p_t - 1) = 1$ for every prime $q \mid R_t$;
- (B) $p_t \not\equiv 1 \pmod{q}$ for every network prime $q \nmid R_t$;
- (C) p_b is a q -th power residue modulo p_t if and only if $\varepsilon_{b,q} = 0$, for all $b < t$ and all $q \mid R_t$;
- (D) $p_t \equiv c_b \pmod{p_b}$ for every $b < t$;
- (E) $p_t \equiv w \pmod{W}$.

Any such p_t satisfies, for every $b < t$,

$$\text{ord}_{p_b}(p_t) = \text{ord}_{p_b}(c_b), \quad \{q \mid R_t : q \mid \text{ord}_{p_t}(p_b)\} = \{q \mid R_t : \varepsilon_{b,q} = 1\}. \quad (3)$$

Proof. Put

$$M_t := W \cdot R^2 \cdot \prod_{b < t} p_b,$$

so that $R_t \mid M_t$ and the three factors are pairwise coprime, the p_b exceeding every network prime and W being coprime to both.

Conditions (A), (B), (D), and (E) are congruence conditions on p_t modulo pairwise coprime divisors of M_t . Each is individually satisfiable: (A) asks, for each $q \mid R_t$, that p_t lie in one of the $q - 1$ classes modulo q^2 that reduce to 1 modulo q but differ from 1 modulo q^2 , and $q - 1 \geq 2$; (B) removes the single class 1 mod q ; (D) and (E) prescribe single classes coprime to their moduli. By the Chinese remainder theorem they jointly define a nonempty union $\mathcal{C}$ of residue classes modulo M_t , that is, a nonempty subset of $\text{Gal}(\mathbb{Q}(\zeta_{M_t})/\mathbb{Q})$. By the first half of (A), every element of $\mathcal{C}$ restricts to the identity on $\mathbb{k}_t = \mathbb{Q}(\zeta_{R_t})$.

Condition (C) prescribes the Frobenius class of p_t in the Kummer compositum F_t of Proposition 5: for p_t unramified in F_t and split in $\mathbb{k}_t$ (the latter being the first half of (A)), the element p_b is a q -th power residue modulo p_t precisely when the Frobenius of p_t fixes $p_b^{1/q}$ [18]. If $t = 1$ there is no such pair (b, q) and (C) is vacuous; assume $t \geq 2$ and let

$$T := \{(b, q) : b < t, q \mid R_t, \varepsilon_{b,q} = 1\},$$

so that (C) requires the Frobenius of p_t to lie in the set $\mathcal{D} := \mathcal{D}_T \subseteq \text{Gal}(F_t/\mathbb{k}_t)$ of Corollary 6. That corollary shows that $\mathcal{D}$ is nonempty and that membership in $\mathcal{D}$ depends only on the conjugacy class of an element, so that (C) is a well-posed condition on the Frobenius class of p_t ; and every element of $\mathcal{D}$ restricts to the identity on $\mathbb{k}_t$, since $\mathcal{D} \subseteq \text{Gal}(F_t/\mathbb{k}_t)$.

Let $L := F_t \cdot \mathbb{Q}(\zeta_{M_t})$. By Proposition 5(3) we have $F_t \cap \mathbb{Q}(\zeta_{M_t}) = \mathbb{k}_t$, so restriction identifies

$$\text{Gal}(L/\mathbb{Q}) \cong \text{Gal}(F_t/\mathbb{Q}) \times_{\text{Gal}(\mathbb{k}_t/\mathbb{Q})} \text{Gal}(\mathbb{Q}(\zeta_{M_t})/\mathbb{Q}).$$

Every element of $\mathcal{C}$ and of $\mathcal{D}$ restricts to the identity on $\mathbb{k}_t$, so the two prescriptions are compatible and the set

$$\mathcal{S} := \{g \in \text{Gal}(L/\mathbb{Q}) : g|_{F_t} \in \mathcal{D}, g|_{\mathbb{Q}(\zeta_{M_t})} \in \mathcal{C}\}$$

is nonempty. We claim that $\mathcal{S}$ is stable under conjugation. Let $g \in \mathcal{S}$ and $h \in \text{Gal}(L/\mathbb{Q})$. Restriction to $\mathbb{Q}(\zeta_{M_t})$ is a homomorphism onto an abelian group, so $(hgh^{-1})|_{\mathbb{Q}(\zeta_{M_t})} = g|_{\mathbb{Q}(\zeta_{M_t})} \in \mathcal{C}$. Restriction to F_t is a homomorphism onto $\text{Gal}(F_t/\mathbb{Q})$, whence $(hgh^{-1})|_{F_t} = h|_{F_t} g|_{F_t} (h|_{F_t})^{-1}$, which lies in $\mathcal{D}$ by Corollary 6.

Thus, $gh^{-1} \in \mathcal{S}$, and $\mathcal{S}$ is a nonempty union of conjugacy classes of $\text{Gal}(L/\mathbb{Q})$. Only finitely many rational primes ramify in L ; discarding those, the Chebotarev density theorem [11, 7] gives a set of positive density of unramified primes whose Frobenius class lies in $\mathcal{S}$. Each of them satisfies (A)–(E), and infinitely many exceed every network prime and every p_b .

For Equation (3): the first identity follows from (D), since $p_t \equiv c_b \pmod{p_b}$. For the second, (A) gives $v_q(p_t - 1) = 1$ for every $q \mid R_t$, so Lemma 4 converts the residue conditions (C) into the divisibility statements $q \mid \text{ord}_{p_t}(p_b)$ if and only if $\varepsilon_{b,q} = 1$. $\square$

Lemma 7 (Column isolation). *Let $p_1, \dots, p_n$ be produced by successive applications of Lemma 6. If q is a network prime with $q \nmid R_j$, then $q \nmid p_j - 1$; consequently $q \nmid r_{ij}$ for every $i \neq j$.*

Proof. Condition (B) at step j gives $p_j \not\equiv 1 \pmod{q}$, that is, $q \nmid p_j - 1$. Since $r_{ij} \mid p_j - 1$ by Fact 1, the second claim follows. $\square$

Theorem 2 (Existence). *Let $n \geq 1$. There exist infinitely many sets $\mathcal{P}$ of n distinct odd primes such that every $e \in I_n$ admits a prime q_e with*

$$q_e \mid r_e \quad \text{and} \quad q_e \nmid r_f \quad \text{for every } f \in I_n \text{ with } f \neq e.$$

Every such set lies in $\mathcal{R}_n$. In particular $\mathcal{R}_n$ is infinite for every $n \geq 1$.

Proof. We first check the second assertion. If q_e is as displayed, then $v_{q_e}(r_e) \geq 1 > 0 = v_{q_e}(r_f)$ for every $f \neq e$, so q_e is a witness prime for e in the sense of Lemma 3; as $e \in I_n$ was arbitrary, that lemma gives $\mathcal{P} \in \mathcal{R}_n$.

For the construction, note first that for $n = 1$ there is no ordered pair of distinct indices and any single odd prime works vacuously. Assume $n \geq 2$ and let $\{q_{ij}\}$ be a network on the full set $J := I_n$.

Select $p_1, \dots, p_n$ by successive applications of Lemma 6, taking $W = 1$ at every step and, at step t :

- for each $b < t$, a class $c_b \in (\mathbb{Z}/p_b\mathbb{Z})^*$ of order exactly q_{tb} , which exists because $q_{tb} \mid R_b \mid p_b - 1$, the latter divisibility holding since R_b is squarefree and (A_b) gives $v_q(p_b - 1) = 1$ for every $q \mid R_b$;
- for each $b < t$ and each $q \mid R_t$, the bit $\varepsilon_{b,q} = 1$ if $q = q_{bt}$ and $\varepsilon_{b,q} = 0$ otherwise.

By Equation (3), the resulting set $\mathcal{P} = \{p_1, \dots, p_n\}$ satisfies, for all $b < t$,

$$r_{tb} = q_{tb}, \quad \{q \mid R_t : q \mid r_{bt}\} = \{q_{bt}\}. \quad (4)$$

In particular, for every ordered pair (i, j) the intended private prime q_{ij} divides r_{ij} , and it is the only prime of R_j that does so.

Fix (i, j) and let $(i', j') \neq (i, j)$ be another ordered pair. If $j' = j$, then $i' \neq i$ and the only prime of R_j dividing $r_{i'j}$ is $q_{i'j} \neq q_{ij}$; since $q_{ij} \mid R_j$, we get $q_{ij} \nmid r_{i'j}$. If $j' \neq j$, then $q_{ij} \mid R_j$ and $\gcd(R_j, R_{j'}) = 1$, so $q_{ij} \nmid R_{j'}$ and Lemma 7 gives $q_{ij} \nmid r_{i'j'}$. Hence, q_{ij} has the stated property, and as (i, j) was arbitrary the construction is complete. It remains to produce infinitely many such sets. For $n = 1$ every odd prime is admissible and the assertion is clear. For $n \geq 2$, keep $p_1, \dots, p_{n-1}$ fixed and apply Lemma 6 once more: it supplies infinitely many admissible primes p_n . The verification above uses only Equation (4) and Lemma 7, and both hold for every admissible choice of p_n , so each such prime yields a set with the stated property; distinct choices yield distinct sets. $\square$

Corollary 7 (Unconditional universality). *For every $n \geq 1$ there exist infinitely many sets $\mathcal{P}$ of n distinct odd primes each of whose families realizes every digraph on the n labelled vertices, that is, $|\Xi(\mathcal{P})| = 2^{n(n-1)}$; equivalently, one for which $\rho(H) > 0$ for every $H \subseteq I_n$.*

Proof. Take $\mathcal{P}$ from Theorem 2 and apply Theorem 1. $\square$

The construction constrains only the network primes. A prime dividing $p_j - 1$ outside the network may well divide r_{ij} , so the orders r_{ij} with $i < j$ need not be prime, and the sets produced are in general far from having all their orders prime. The orders r_{ij} with $j < i$, on the other hand, are prescribed exactly by Equation (4) and are prime; only the $\binom{n}{2}$ pairs above the diagonal escape that control. This asymmetry is the reason the same machinery does not settle the questions of Section 4.3.

4. Beyond Realization

Lcm-independence is the condition under which every digraph occurs. It says nothing about how the occurrences are distributed, and nothing about whether distinct divisors of the fundamental exponent produce distinct digraphs. Two strictly stronger conditions govern those two questions, and this section treats them in turn.

4.1. Pairwise Coprime Sets

Definition 7 (Pairwise coprime set). The set $\mathcal{P}$ is a *pairwise coprime set*, written $\mathcal{P} \in \mathcal{R}_n^c$, if $\mathcal{P} \in \mathcal{R}_n$ and

$$\gcd(r_e, r_f) = 1 \quad \text{for all } e, f \in I_n \text{ with } e \neq f.$$

The requirement that $\mathcal{P}$ lie in $\mathcal{R}_n$ makes $\mathcal{R}_n^c$ a class of realizing sets by construction, but it is not an independent hypothesis: given the coprimality of the orders, that requirement amounts to the exclusion of the trivial order 1.

Proposition 6. *The set $\mathcal{P}$ lies in $\mathcal{R}_n^c$ if and only if $r_e \geq 2$ for every $e \in I_n$ and $\gcd(r_e, r_f) = 1$ for all $e \neq f$. The inclusion $\mathcal{R}_n^c \subseteq \mathcal{R}_n$ is strict for $n = 3$.*

Proof. Suppose $\mathcal{P} \in \mathcal{R}_n^c$ and $r_e = 1$ for some e . Then r_e divides $\text{lcm}\{r_f : f \neq e\}$, contradicting lcm-independence, so $r_e \geq 2$ for every e ; coprimality is part of the definition.

Conversely, assume the two stated conditions and fix $e \in I_n$. Since $r_e \geq 2$ there is a prime q with $q \mid r_e$. If $q \mid r_f$ for some $f \neq e$, then $q \mid \gcd(r_e, r_f) = 1$, which is absurd; hence $v_q(r_e) \geq 1 > 0 = v_q(r_f)$ for every $f \neq e$, and Lemma 3 gives $\mathcal{P} \in \mathcal{R}_n$. With coprimality this places $\mathcal{P}$ in $\mathcal{R}_n^c$.

For strictness, Example 3 exhibits a member of $\mathcal{R}_3$ with two orders sharing a factor. $\square$

Example 3. Let $\mathcal{P} = \{1013, 1021, 403649\}$, indexed in this order. Its orders are

$$\begin{aligned} r_{12} &= 2^2 \cdot 5 \cdot 17, & r_{13} &= 2^5 \cdot 7 \cdot 17, & r_{23} &= 2 \cdot 17 \cdot 53, \\ r_{21} &= 2^2 \cdot 23, & r_{31} &= 2 \cdot 11, & r_{32} &= 3 \cdot 17. \end{aligned}$$

Each order carries a prime that no other carries (respectively 5, 7, 53, 23, 11, and 3), so $\mathcal{P} \in \mathcal{R}_3$. But $\gcd(r_{12}, r_{13}) = 68$, so $\mathcal{P} \notin \mathcal{R}_3^c$.

The interest of $\mathcal{R}_n^c$ is that it is exactly the class on which the realization density of Definition 4 becomes a product measure.

Theorem 3 (Independence of the edges). *Let $\mathcal{P} \in \mathcal{R}_n$. The following are equivalent.*

1. *we have $\mathcal{P} \in \mathcal{R}_n^c$;*
2. *the fundamental exponent satisfies $m_{\max}(\mathcal{P}) = \prod_{e \in I_n} r_e$;*
3. *for every $H \subseteq I_n$,*

$$\rho(H) = \prod_{e \in H} \frac{1}{r_e} \prod_{e \notin H} \left(1 - \frac{1}{r_e}\right). \quad (5)$$

When these hold, $m_H = \prod_{e \in H} r_e$ for every $H \subseteq I_n$, and under ρ the edge indicators are independent random variables, the edge e being present with probability $1/r_e$.

Proof. Condition (1) implies (2), together with the supplementary identity: for pairwise coprime integers the least common multiple is the product, so $m_H = \text{lcm}\{r_e : e \in H\} = \prod_{e \in H} r_e$ for every H , and $H = I_n$ is (2).

Condition (2) implies (1). In general $\text{lcm}\{r_e\} \leq \prod_e r_e$, with equality only if the r_e are pairwise coprime: if $g := \gcd(r_e, r_f) > 1$ for some $e \neq f$, then $\text{lcm}\{r_e\}$ divides $(\prod_e r_e)/g$, which is strictly smaller.

Condition (1) implies (3). Write $H' = H \sqcup H_0$ with $H_0 \subseteq I_n \setminus H$. By the identity just proved, $m_{H'} = \prod_{e \in H} r_e \cdot \prod_{e \in H_0} r_e$, so Equation (1) becomes

$$\begin{aligned} \rho(H) &= \sum_{H_0 \subseteq I_n \setminus H} \frac{(-1)^{|H_0|}}{\prod_{e \in H} r_e \cdot \prod_{e \in H_0} r_e} \\ &= \left(\prod_{e \in H} \frac{1}{r_e} \right) \sum_{H_0} \prod_{e \in H_0} \left(\frac{-1}{r_e} \right) \\ &= \left(\prod_{e \in H} \frac{1}{r_e} \right) \prod_{e \notin H} \left(1 - \frac{1}{r_e} \right), \end{aligned}$$

the last step being the expansion of a product of binomials over the subsets H_0 of $I_n \setminus H$.

Condition (3) implies (2). Take $H = I_n$ in Equation (5), so that the second product is empty and $\rho(I_n) = \prod_e r_e^{-1}$. On the other hand $\{m : E_m(\mathcal{P}) = I_n\}$ is the set of multiples of $m_{\max}(\mathcal{P})$ by Corollary 1(3), of density $1/m_{\max}(\mathcal{P})$. Comparing gives (2).

For the final assertion, Equation (5) is precisely the statement that under ρ the events $\{e \in E_m(\mathcal{P})\}$, $e \in I_n$, are independent with the stated probabilities. $\square$

Outside $\mathcal{R}_n^c$ a prime shared by two orders correlates the corresponding edges and the measure does not factor; Theorem 3 says that this is the only obstruction. Coprimality also forces a severe constraint on parity, since every order divides an even number.

Proposition 7 (Parity obstruction). *Let $\mathcal{P} \in \mathcal{R}_n^c$. Then at most one of the $n(n-1)$ orders r_e is even.*

Proof. Two even orders would share the factor 2, contradicting $\gcd(r_e, r_f) = 1$. $\square$

Lemma 8 (Residue form). *Let $\mathcal{P}$ be any set of n distinct odd primes and let $(i, j) \in I_n$. Then p_i generates the unique subgroup of order r_{ij} of $(\mathbb{Z}/p_j\mathbb{Z})^*$, and consequently p_i is an s -th power residue modulo p_j for every divisor s of $(p_j-1)/r_{ij}$. In particular:*

1. *if r_{ij} is odd, then $\left(\frac{p_i}{p_j}\right) = 1$;*
2. *if $r_{ij} = 2$, then $p_i \equiv -1 \pmod{p_j}$, so that $p_j \mid p_i + 1$.*

Proof. The group $(\mathbb{Z}/p_j\mathbb{Z})^*$ is cyclic of order $p_j - 1$, so it contains exactly one subgroup of each order dividing $p_j - 1$; since p_i has order exactly r_{ij} , it generates that subgroup. If $s \mid (p_j - 1)/r_{ij}$, then $r_{ij} \mid (p_j - 1)/s$, so p_i lies in the subgroup of index s , which is precisely the set of s -th powers.

For (1), $p_j - 1$ is even and r_{ij} is odd, so $2 \mid (p_j - 1)/r_{ij}$ and p_i is a square. For (2), $p_i^2 \equiv 1$ and $p_i \not\equiv 1 \pmod{p_j}$, whence $p_i \equiv -1 \pmod{p_j}$. $\square$

Theorem 4 (Parity rigidity). *Let $n \geq 2$ and let $\mathcal{P} \in \mathcal{R}_n^c$. Then*

$$|\{p \in \mathcal{P} : p \equiv 3 \pmod{4}\}| \leq 2.$$

If equality holds, say $p_a \equiv p_b \equiv 3 \pmod{4}$ with $a \neq b$, then after possibly exchanging a and b the order r_{ab} is the unique even one among the $n(n-1)$ orders, and $\left(\frac{p_i}{p_j}\right) = 1$ for every $(i, j) \in I_n$ with $(i, j) \neq (a, b)$. In particular, for $n \geq 3$ at least $n-2$ elements of $\mathcal{P}$ are congruent to 1 modulo 4.

Proof. Let k denote the number of elements of $\mathcal{P}$ congruent to 3 modulo 4, and let $\{a, b\}$ be an unordered pair of such elements. Quadratic reciprocity gives

$$\left(\frac{p_a}{p_b}\right)\left(\frac{p_b}{p_a}\right) = (-1)^{\frac{p_a-1}{2} \cdot \frac{p_b-1}{2}} = -1,$$

so at least one of the two Legendre symbols equals -1 ; by Lemma 8(1) the corresponding order, r_{ab} or r_{ba} , is then even. Distinct unordered pairs of elements of $\mathcal{P}$ give rise to disjoint sets of ordered pairs in I_n , so distinct unordered pairs produce distinct ordered pairs carrying an even order. By Proposition 7 there is at most one such ordered pair, hence at most one such unordered pair, that is $\binom{k}{2} \leq 1$ and $k \leq 2$.

Assume $k = 2$, with $p_a \equiv p_b \equiv 3 \pmod{4}$. By the previous paragraph one of r_{ab}, r_{ba} is even; relabel so that r_{ab} is, and by Proposition 7 it is the only even order. Every pair $(i, j) \neq (a, b)$ therefore has odd order, and Lemma 8(1) gives $\left(\frac{p_i}{p_j}\right) = 1$. The final assertion is the case $n \geq 3$ of $k \leq 2$. $\square$

Example 3 shows how far a realizing set may sit from this regime: five of its six orders are even. In a member of $\mathcal{R}_n^c$, by contrast, Lemma 8(1) and Proposition 7 together leave at most one ordered pair $(i, j) \in I_n$ with $\left(\frac{p_i}{p_j}\right) = -1$; every element of $\mathcal{P}$ is a quadratic residue modulo every other, with at most one exception among the $n(n-1)$ ordered pairs.

4.2. Aligned Sets

Definition 8 (Aligned set). The set $\mathcal{P}$ is an *aligned set*, written $\mathcal{P} \in \mathcal{R}_n^a$, if $\mathcal{P} \in \mathcal{R}_n$ and the $n(n-1)$ orders r_e are prime and pairwise distinct.

Pairwise distinct primes are pairwise coprime and exceed 1, so by Proposition 6 the realizing hypothesis is again automatic, and $\mathcal{R}_n^a \subseteq \mathcal{R}_n^c$. The inclusion is strict for $n = 2$: the set $\{3, 71\}$ has $\text{ord}_{71}(3) = 35$ and $\text{ord}_3(71) = 2$, coprime but not both prime.

Theorem 5 (Faithfulness). *Write $m_{\max} := m_{\max}(\mathcal{P})$ and let*

$$\Psi : \{m' \in \mathbb{N} : m' \mid m_{\max}\} \longrightarrow 2^{I_n}, \quad \Psi(m') := E_{m'}(\mathcal{P}).$$

The following are equivalent.

1. we have $\mathcal{P} \in \mathcal{R}_n^a$;
2. the family satisfies $|\Xi(\mathcal{P})| = d(m_{\max}) = 2^{n(n-1)}$, that is, both bounds of Proposition 1 are attained;
3. the map Ψ is an isomorphism of Boolean algebras from the divisor lattice of $m_{\max}$ onto the power set of I_n .

Proof. Condition (2) implies (1). By Theorem 1 the hypothesis $|\Xi(\mathcal{P})| = 2^{n(n-1)}$ gives $\mathcal{P} \in \mathcal{R}_n$, and by Corollary 1(2) the map Ψ is onto $\Xi(\mathcal{P}) = 2^{I_n}$. Its domain has $d(m_{\max}) = 2^{n(n-1)}$ elements, so Ψ is a bijection, and every divisor of $m_{\max}$ equals m_H for exactly one H , by Lemma 2.

Suppose $q^2 \mid m_{\max}$ for some prime q . Then q and q^2 are divisors of $m_{\max}$, so $q = m_H$ and $q^2 = m_{H'}$ for suitable H, H' . Neither is empty, since $m_{\emptyset} = 1$; and a least common multiple equal to q , all of whose terms divide q , forces $r_e = q$ for some $e \in H$, while one equal to q^2 forces $r_f = q^2$ for some $f \in H'$. Then $e \neq f$ and $r_e \mid r_f$, so r_e divides $\text{lcm}\{r_g : g \neq e\}$, contradicting lcm-independence. Hence, $m_{\max}$ is squarefree, and $d(m_{\max}) = 2^{\omega(m_{\max})} = 2^{n(n-1)}$ gives $\omega(m_{\max}) = n(n-1)$.

By Proposition 4 there are $n(n-1)$ pairwise distinct witness primes q_e , each dividing $m_{\max}$; as $m_{\max}$ has exactly $n(n-1)$ prime divisors, these are all of them. Fix e and let q be any prime dividing r_e ; then $q = q_f$ for some f . Since $m_{\max}$ is squarefree, so is r_e , and all valuations involved lie in $\{0, 1\}$; the defining inequality $v_{q_f}(r_f) > v_{q_f}(r_g)$ for $g \neq f$ therefore forces $q_f \nmid r_g$ for every $g \neq f$, so $f = e$. Thus, q_e is the only prime dividing r_e , and r_e being squarefree gives $r_e = q_e$. The q_e are pairwise distinct, so $\mathcal{P} \in \mathcal{R}_n^a$.

Condition (1) implies (3). Write $r_e = q_e$ with the q_e pairwise distinct primes. Then $m_{\max} = \prod_{e \in I_n} q_e$ is squarefree with exactly $n(n-1)$ prime factors, so its divisors are the products $\prod_{e \in H} q_e$ over $H \subseteq I_n$, each arising from exactly one H . For such a divisor m' we have $\Psi(m') = \{e : q_e \mid m'\} = H$ by Fact 1. Hence, Ψ is the canonical bijection between divisors and subsets, and it carries gcd to intersection, lcm to union, 1 to $\emptyset$, and $m_{\max}$ to I_n ; that is, it is an isomorphism of Boolean algebras.

Condition (3) implies (2). A bijection forces $d(m_{\max}) = |2^{I_n}| = 2^{n(n-1)}$, and its image is $\Xi(\mathcal{P})$ by Corollary 1(2), so $|\Xi(\mathcal{P})| = 2^{n(n-1)}$ as well. $\square$

Corollary 8 (The arithmetic reads the digraphs). *Let $\mathcal{P} \in \mathcal{R}_n^a$ and let A, B divide $m_{\max}(\mathcal{P})$. Then:*

1. we have $E_A(\mathcal{P}) \subseteq E_B(\mathcal{P})$ if and only if $A \mid B$; in particular $E_A(\mathcal{P}) = \emptyset$ if and only if $A = 1$;
2. we have $m_{H \sqcup H'} = m_H \cdot m_{H'}$ for all disjoint $H, H' \subseteq I_n$.

Proof. Both assertions are the transport along the isomorphism Ψ of Theorem 5(3) of the corresponding statements about subsets. Explicitly, with $r_e = q_e$ pairwise distinct primes, a divisor A of the squarefree integer $m_{\max}$ is the product of the q_e with $e \in E_A(\mathcal{P})$; assertion (1) is then the equivalence of $E_A \subseteq E_B$ with divisibility of squarefree integers, and $A = 1$ corresponds to the empty set. For (2), $m_H = \prod_{e \in H} q_e$ and the two index sets are disjoint. $\square$

Alignment supplies the union counterpart of Corollary 2.

Corollary 9 (Union of edge sets). *Let $\mathcal{P} \in \mathcal{R}_n^a$. Then, for all $m, m' \in \mathbb{N}$,*

$$E_m(\mathcal{P}) \cup E_{m'}(\mathcal{P}) = E_{\text{lcm}(m, m')}(\mathcal{P}).$$

Proof. Write $r_e = q_e$ with the q_e pairwise distinct primes, as in Definition 8, and fix $e \in I_n$. By Fact 1, the pair e lies in $E_{\text{lcm}(m, m')}(\mathcal{P})$ if and only if $q_e \mid \text{lcm}(m, m')$. The valuation of a least common multiple is the maximum of the valuations, so $v_{q_e}(\text{lcm}(m, m')) \geq 1$ if and only if $v_{q_e}(m) \geq 1$ or $v_{q_e}(m') \geq 1$; as q_e is prime, this says that q_e divides m or q_e divides m' , that is, that e lies in $E_m(\mathcal{P})$ or in $E_{m'}(\mathcal{P})$. $\square$

Neither conclusion survives the weakening from alignment to coprimality of the orders. For $\mathcal{P} = \{11, 37\}$, which lies in $\mathcal{R}_2^c$ because its orders 6 and 5 are coprime, the divisors $A = 2$ and $B = 3$ of $m_{\max}(\mathcal{P}) = 30$ satisfy $E_A(\mathcal{P}) = E_B(\mathcal{P}) = \emptyset$ while $A \nmid B$, and $E_A(\mathcal{P}) \cup E_B(\mathcal{P}) = \emptyset$ while $E_{\text{lcm}(A, B)}(\mathcal{P}) = \{(1, 2)\}$. Both failures are elementary consequences of the presence of a composite order.

Corollary 10 (Parity rigidity, aligned case). *Let $n \geq 2$ and let $\mathcal{P} \in \mathcal{R}_n^a$. Then at most two elements of $\mathcal{P}$ are congruent to 3 modulo 4, and if exactly two, say $p_a \equiv p_b \equiv 3 \pmod{4}$, then after possibly exchanging a and b :*

1. *we have $r_{ab} = 2$, and (a, b) is the unique pair of I_n whose order is even;*
2. *we have $p_b \mid p_a + 1$;*
3. *the order r_{ij} is an odd prime and $\left(\frac{p_i}{p_j}\right) = 1$ for every $(i, j) \neq (a, b)$.*

Proof. As $\mathcal{R}_n^a \subseteq \mathcal{R}_n^c$, Theorem 4 applies and yields the bound, the uniqueness of the even order r_{ab} , and the Legendre symbols in (3). In $\mathcal{R}_n^a$ every order is prime, so an even order equals 2, which is (1), and the remaining orders are odd primes. Assertion (2) is Lemma 8(2) applied to the pair (a, b) . $\square$

Example 4. Let $\mathcal{P} = \{3, 11\}$, indexed in this order. Then $r_{12} = \text{ord}_{11}(3) = 5$ and $r_{21} = \text{ord}_3(11) = 2$, two distinct primes, so $\mathcal{P} \in \mathcal{R}_2^a$. Here $m_{\max}(\mathcal{P}) = 10$ has $d(10) = 4 = 2^2$ divisors, matching Theorem 5(2). Both elements are congruent to 3 modulo 4, so the bound of Corollary 10 is attained; the even order is $r_{21} = 2$ and indeed $3 \mid 11 + 1$.

4.3. Constraints and the Non-Existence Conjecture

No aligned set of cardinality $n \geq 3$ is known. This subsection collects the unconditional constraints that such a set would have to satisfy, records the count they support, and states the resulting conjecture. Both classes of this section are closed under passage to subsets.

Proposition 8 (Heredity). *Let $\mathcal{P}' \subseteq \mathcal{P}$ with $|\mathcal{P}'| = s$. If $\mathcal{P} \in \mathcal{R}_n^c$ then $\mathcal{P}' \in \mathcal{R}_s^c$, and if $\mathcal{P} \in \mathcal{R}_n^a$ then $\mathcal{P}' \in \mathcal{R}_s^a$.*

Proof. By Proposition 3 we have $\mathcal{P}' \in \mathcal{R}_s$. The order r_{ij} depends only on the ordered pair (p_i, p_j) and not on the ambient set, so the orders of $\mathcal{P}'$ form a subfamily of those of $\mathcal{P}$; being pairwise coprime in one case, and prime and pairwise distinct in the other, are conditions inherited by subfamilies. $\square$

Corollary 11. *If $\mathcal{R}_s^a = \emptyset$ for some $s \geq 3$, then $\mathcal{R}_n^a = \emptyset$ for every $n \geq s$. The same implication holds with $\mathcal{R}^c$ in place of $\mathcal{R}^a$.*

Proof. If $\mathcal{P} \in \mathcal{R}_n^a$ with $n \geq s$, any subset of cardinality s lies in $\mathcal{R}_s^a$ by Proposition 8. $\square$

The first constraint is on the size of the elements: alignment forces each $p_j - 1$ to be divisible by a product of prescribed distinct primes, not merely by $n - 1$ of them.

Proposition 9 (Capacity of an aligned set). *Let $\mathcal{P} \in \mathcal{R}_n^a$ with $n \geq 2$. Then*

$$\prod_{i \neq j} r_{ij} \mid p_j - 1 \quad \text{for every } j,$$

so that $\omega(p_j - 1) \geq n - 1$; and $\omega(p_j - 1) \geq n$ for every j with at most one exception.

Proof. Each r_{ij} divides $p_j - 1$ by Fact 1, and the $n - 1$ orders r_{ij} with $i \neq j$ are pairwise distinct primes by Definition 8; a set of pairwise distinct primes each dividing $p_j - 1$ has product dividing $p_j - 1$, whence the displayed divisibility, recovering $\omega(p_j - 1) \geq n - 1$ of Corollary 3. If none of those orders equals 2 they are all odd, while $2 \mid p_j - 1$, so $p_j - 1$ carries the additional prime factor 2 and $\omega(p_j - 1) \geq n$. Since $\mathcal{R}_n^a \subseteq \mathcal{R}_n^c$, Proposition 7 allows at most one of the $n(n - 1)$ orders to be even, hence at most one to equal 2, and so at most one index j admits the exception. $\square$

The second constraint is of a different nature, and explains why the literature discussed in the introduction does not bear on the present question. Prescribing an order exactly, rather than up to divisibility, replaces a union of residue classes by the divisor set of a single integer.

Theorem 6 (Cyclotomic confinement). *Let r be a prime, let p be a prime with $p \neq r$, and let Φ_r denote the r -th cyclotomic polynomial. For every prime $q \notin \{p, r\}$,*

$$\text{ord}_q(p) = r \quad \text{if and only if} \quad q \mid \Phi_r(p).$$

Consequently, let p_1 be a prime and let $r_2, \dots, r_n$ be primes. The aligned sets $\mathcal{P} = \{p_1, \dots, p_n\} \in \mathcal{R}_n^a$ containing p_1 and indexed so that $r_{1j} = r_j$ for $2 \leq j \leq n$ are finite in number, at most $\prod_{j=2}^n \omega(\Phi_{r_j}(p_1))$ of them, and they are effectively computable from p_1 and $r_2, \dots, r_n$.

Proof. Suppose $\text{ord}_q(p) = r$. Then $q \mid p^r - 1$ while $q \nmid p - 1$, and since r is prime, $\Phi_r(p) = (p^r - 1)/(p - 1)$; hence $q \mid \Phi_r(p)$. Conversely, suppose $q \mid \Phi_r(p)$. Then $q \mid p^r - 1$, so $\text{ord}_q(p)$ divides the prime r and equals 1 or r . Were it 1 we would have $p \equiv 1 \pmod{q}$ and therefore $\Phi_r(p) = 1 + p + \dots + p^{r-1} \equiv r \pmod{q}$, giving $q \mid r$ and contradicting $q \neq r$. So $\text{ord}_q(p) = r$.

For the second assertion, each p_j with $j \geq 2$ satisfies $\text{ord}_{p_j}(p_1) = r_j$, and $p_j \notin \{p_1, r_j\}$ because $r_j = \text{ord}_{p_j}(p_1)$ divides $p_j - 1$ and is therefore smaller than p_j . By the equivalence just proved, p_j lies among the $\omega(\Phi_{r_j}(p_1))$ prime divisors of the fixed integer $\Phi_{r_j}(p_1)$. That integer is explicit and smaller than $p_1^{r_j}$, so the list of candidates is computable, and the product of the counts bounds the number of resulting sets. $\square$

Theorem 6 is the reason no density statement is available for $\mathcal{R}_n^a$. In the direction $i \rightarrow j$ a prescribed exact order confines p_j to the prime divisors of one explicit integer, a set of size $O(\log p_1)$ for fixed r , rather than to a union of residue classes of positive density; the arguments of Schinzel and Wójcik [14], Pappalardi and Susa [12], and Järvinen [5] do not apply. That the confinement is not vacuous is the theorem of Bang [1] and Zsigmondy [20]: for all integers $a \geq 2$ and $k \geq 1$ the integer $a^k - 1$ has a primitive prime divisor, except when $k = 1$ and $a = 2$, when $k = 6$ and $a = 2$, and when $k = 2$ and $a + 1$ is a power of two. Here $a = p$ is an odd prime and $k = r$ is a prime, so the first two exceptions cannot occur and the third leaves only the case $r = 2$ with $p + 1$ a power of two; outside that case there is a prime q with $\text{ord}_q(p) = r$. Alignment is in this respect a primitive-divisor problem rather than an Artin-type problem.

The two constraints bear on the columns of $\mathcal{P}$, and the following count records how restrictive a single column is.

Lemma 9 (Column count). *Let p be an odd prime and let $k \geq 1$. The number of k -tuples $(a_1, \dots, a_k) \in ((\mathbb{Z}/p\mathbb{Z})^*)^k$ whose orders are pairwise distinct primes is less than $\omega(p - 1)^k (p - 1)$, so that the proportion of such k -tuples is less than $\omega(p - 1)^k / (p - 1)^{k-1}$.*

Proof. Fix an ordered k -tuple $(q_1, \dots, q_k)$ of pairwise distinct primes dividing $p - 1$; there are fewer than $\omega(p - 1)^k$ of them. The group $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic of order $p - 1$,

so it contains exactly $q_i - 1$ elements of order q_i , and the number of k -tuples with $\text{ord}(a_i) = q_i$ for each i is $\prod_i (q_i - 1)$. The q_i are pairwise distinct primes dividing $p - 1$, so $\prod_i q_i$ divides $p - 1$ and $\prod_i (q_i - 1) < \prod_i q_i \leq p - 1$. Summing over the tuples of orders gives the first bound, and dividing by $(p - 1)^k$ the second. $\square$

Lemma 9 applies to each of the n columns of a putative aligned set, with $k = n - 1$. Model the $n - 1$ residues $p_i \bmod p_j$ as independent and uniform in $(\mathbb{Z}/p_j\mathbb{Z})^*$, independently across j ; this ignores the constraints of Proposition 9 and Theorem 6, both of which only reduce the count further, as well as the requirement that the $n(n - 1)$ orders be distinct across columns. Under that model the expected number of aligned sets with all elements at most x is at most

$$\frac{1}{n!} \left(\sum_{p \leq x} \frac{\omega(p-1)^{n-1}}{(p-1)^{n-2}} \right)^n,$$

and $\omega(p-1) < \log_2 p$. For $n \geq 4$ the exponent $n-2$ is at least 2, the series converges, and the bound remains finite as $x \rightarrow \infty$. Its value can be estimated. Summing the terms over the odd primes below $2 \cdot 10^6$ with $n = 4$ gives less than 0.856, and the remaining terms are bounded, using $\omega(p-1) < \log_2 p$ and the monotonicity of $t \mapsto (\log_2 t)^3/(t-1)^2$, by

$$\int_{2 \cdot 10^6}^{\infty} \frac{(\log_2 t)^3}{(t-1)^2} dt < 0.006.$$

The whole series is therefore less than 0.861, and the bound for $n = 4$ is less than 0.023; the corresponding bounds for $n \geq 5$ are smaller by several orders of magnitude, their total being below 10^{-5} , so the bound for all $n \geq 4$ together remains less than 0.023. Under the model, in other words, aligned sets of cardinality at least 4 are expected not to occur at all. For $n = 3$ the series is $\sum_p (\log_2 p)^2/(p-1)$, which diverges, and the bound is vacuous.

The model overestimates substantially. Truncated at $x = 3 \cdot 10^5$, the same expression with $n = 3$ evaluates to more than 400, whereas by Proposition 10 the true number of aligned sets in that range is zero. The figure obtained for $n = 4$ should accordingly be read as an upper bound for a heuristic count rather than as a prediction, and the true expectation is presumably far smaller.

The constraints just proved are also open to direct verification. Two searches were carried out, and both are described below in enough detail to be repeated.

Proposition 10 (Verification). *The following two statements hold.*

1. *there is no aligned set of cardinality 3 all of whose elements are smaller than $3 \cdot 10^5$, and hence none of cardinality $n \geq 3$ with that property.*
2. *let p be a prime with $p < 100$ and let $r \neq r'$ be primes with $r, r' \leq 23$. Then there is no aligned set $\{p, p', p''\}$ with $\text{ord}_{p'}(p) = r$ and $\text{ord}_{p''}(p) = r'$; here p' and p'' are subject to no upper bound.*

Proof. For (1), let G be the graph whose vertices are the odd primes below $3 \cdot 10^5$, with p and p' adjacent when $\text{ord}_{p'}(p)$ and $\text{ord}_p(p')$ are both prime and distinct from each other. By Definition 8 the elements of an aligned set of cardinality 3 are pairwise adjacent in G , so it suffices to inspect the triangles of G ; the second assertion of (1) then follows from Proposition 8, an aligned set of cardinality $n \geq 3$ containing one of cardinality 3. The order of p modulo p' is prime if and only if

$$p \not\equiv 1 \pmod{p'} \quad \text{and} \quad p^q \equiv 1 \pmod{p'} \quad \text{for some prime } q \mid p' - 1,$$

and exactly one prime q then satisfies the second condition, namely $q = \text{ord}_{p'}(p)$. Indeed, if the order is a prime q then $q \mid p' - 1$ by Fact 1 and both conditions hold; conversely the second gives $\text{ord}_{p'}(p) \mid q$, so the order is 1 or q , and the first excludes 1. Two distinct primes q, q' satisfying the second condition would give $\text{ord}_{p'}(p) \mid \gcd(q, q') = 1$, excluded again by the first. The 25 996 odd primes below $3 \cdot 10^5$ form 675 766 020 ordered pairs, of which 42 465 610 have prime order; these give 1 237 384 edges of G . Its 8 877 957 triangles were examined; none has its six orders pairwise distinct, and 89 have exactly five distinct orders. The factorizations of $p - 1$ for all p below the bound were obtained in a single pass by a sieve recording the smallest prime factor of each integer, rather than one integer at a time.

For (2), fix p and r . By Theorem 6 a prime p' with $\text{ord}_{p'}(p) = r$ is a prime divisor of $\Phi_r(p)$, so the complete list of candidates is obtained by factoring that single integer, and no bound on p' is involved. For every odd prime $p < 100$ and every prime $r \leq 23$ the integer $\Phi_r(p)$ has at most 45 decimal digits and was factored completely. The factorizations were carried out with the general-purpose routine `factorint` of the `SymPy` library, version 1.14.0. Pairing the resulting candidates gave 3949 unordered pairs (p', p'') , and for each of them the remaining four orders were computed; in no case are the six orders prime and pairwise distinct.

Every order reported by either search was recomputed with an independent routine, with no discrepancy. $\square$

Code implementing both searches, together with the resulting data, is archived at Zenodo [15].

Example 5. Let $\mathcal{P} = \{11, 43, 16333\}$, indexed in this order. Its orders are

$$r_{12} = 7, \quad r_{13} = 1361, \quad r_{21} = 2, \quad r_{23} = 1361, \quad r_{31} = 5, \quad r_{32} = 3,$$

six primes, five of them distinct. The repetition is $r_{13} = r_{23}$, the two orders modulo the same element 16333; here $16333 - 1 = 2^2 \cdot 3 \cdot 1361$, so $\omega(16333 - 1) = 3$ and the two orders entering that element have only three primes to choose between. Among the 89 sets found in the proof of Proposition 10(1) with five distinct orders, every repetition is of this shape: two orders modulo a common element.

We state the conjecture only in the range where the count has content.

Conjecture 1. The class $\mathcal{R}_n^a$ is empty for every $n \geq 4$.

By Corollary 11 the conjecture reduces to the single case $n = 4$. The case $n = 3$ is left open, and we emphasize that the estimate above is silent on it rather than adverse to it: a divergent upper bound carries no prediction either way.

Question 1. Is $\mathcal{R}_3^a$ empty?

An affirmative answer implies Conjecture 1, again by Corollary 11; a negative one leaves it untouched, since alignment at cardinality 3 does not propagate upwards. A second question sits below both, and a negative answer to it would settle Conjecture 1 under a weaker hypothesis, coprimality of the orders rather than primality.

Question 2. Is $\mathcal{R}_n^c$ nonempty for some $n \geq 3$?

A positive answer would isolate the difficulty in the primality of the orders rather than in their independence: by Theorem 3 a member of $\mathcal{R}_n^c$ already carries the product measure, and alignment adds to that only the requirement that each order be prime, which by Theorem 5 is what turns Ψ into an isomorphism of Boolean algebras.

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